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05-12-2023

Original Article Title: “Electricity Theft Detection for Smart Homes: Harnessing the Power of Machine Learning with Real and Synthetic Attacks”

To: IEEE Access Editor

Re: Response to reviewers

Dear Editor,

Thank you for allowing a resubmission of our manuscript, with an opportunity to address the reviewers' comments.

We are uploading (a) our point-by-point response to the comments (below) (response to reviewers, under “Author’s Response Files”), (b) an updated manuscript with yellow highlighting indicating changes (as “Highlighted PDF”), and (c) a clean updated manuscript without highlights (“Main Manuscript”).

Best regards,

Abraham et al.

Reviewer#1, Concern # 1 (please list here): No comment

Author response: No comment

Author action: We updated the manuscript by

Reviewer#1, Concern # 2 (please list here): No comment

Author response: No comment

Author action: We updated the manuscript by

Reviewer#1, Concern # 3 (please list here): No comment

Author response: No comment

Author action: We updated the manuscript by

Reviewer#2, Concern # 1 (please list here): No comment

Author response: No comment

Author action: We updated the manuscript by

Reviewer#2, Concern # 2 (please list here): No comment

Author response: No comment

Author action: We updated the manuscript by

Reviewer#2, Concern # 3 (please list here): No comment

Author response: No comment

Author action: We updated the manuscript by

Reviewer#3, Concern # 1 (please list here): 1. The main motivation of the paper as well as a summary of the contribution should be stated in the abstract.

Author response: Thank you for taking your precious time to read and review our manuscript. We appreciate your valuable suggestions and have revised our paper accordingly. Below are the responses to your questions and concerns:

Abstract re-formatted: Addressing the critical issue of electricity theft in smart power grids, which causes significant economic losses to both homeowners and utility companies, this paper introduces a novel machine learning (ML)-based framework for its detection in smart homes. Departing from traditional detection methods like manual inspections, our framework leverages ML to analyze variable electric consumption patterns influenced by lifestyle and weather. We utilize knowledge-based synthetic attack data (KBSAD), encompassing both benign and malicious patterns, to train an attack classifier that simulates real-world scenarios. The framework was validated using the Almanac of Minutely Power dataset version 2 (AMPds2), preprocessed for binary classification to detect anomalies and evaluated with real attack dataset. Our findings demonstrate the exceptional performance of the Extreme Gradient Boosting binary-based discriminator algorithm, achieving average AUC scores of 98.74% and 98.69% for real and simulated attacks across various homes. This method notably surpassed legacy unsupervised techniques and other synthetic binary discriminators (SYN)such as Random Forest (RF) and Multi-Layer Perceptron (MLP) algorithms. By employing KBSAD, our approach eliminates the need for extensive real attack data collection and seamlessly integrates synthetic attacks with legitimate meter readings, marking a notable advancement in smart home electricity theft detection (ETD).

Author action: We updated the manuscript by reformatting the abstract to include main motivation of the paper as well as summary of the contribution

Reviewer#3, Concern # 2 (please list here): 2. It is better to divide the introduction into the following sections:

- a. The motivation of the paper:
- b. Literature Review:
- c. Study gaps:
- d. Contribution:

Author response: (a) Motivation: Our focus is on securing homes from the threat of electricity theft by external parties. In cases where an attacker steals electricity not directly from the power grid but from neighboring houses, the utility company may remain indifferent to such theft, leaving homeowners responsible for safeguarding their own resources.

(b) Literature Review:

Reference	Method	Contributions	Limitations	Our Model's Resolution
Punmiya et al., 2019 [16]	Simple statistical models: random, mean, shift, reverse for synthetic attack loads	These models provided initial frameworks for synthetic load generation and attack simulation.	Limited complexity and lack of realism in attack scenarios.	Introduces a novel set of attack scenarios (Baseload, Weakload, Peakhour, Midnight, Evil-Twin) that simulate more realistic electricity theft patterns.

Abraham et al., 2023 [17]	Knowledge-based synthetic attack generation	Rooted in understanding the intentions behind various attacks.	Lack empirical validation with real attack data.	Validates the extended model with real attack data for binary classification and anomaly detection.
Dayaratne et al., 2021 [18]; Singh et al., 2017 [19]; Punmiya et al., 2019 [16]	Protection of utility companies from electricity theft	Focus on large-scale protection to prevent revenue loss from power stealing.	Issue of power theft between individual homes are not addressed.	Our model enables households to autonomously detect and protect against electricity theft, including scenarios where power is stolen from one home by another.

This table encapsulates the research landscape, showing how each piece of literature contributes to the field of Electricity Theft Detection (ETD) while also highlighting the gaps our proposed model aims to fill. Our model not only adopts more sophisticated attack simulations but also extends validation methods to include real-world attack data, thereby enhancing the practical application of ETD systems in smart homes.

(c) Study Gaps:

- **Limited Reduction Techniques:** Prior research, such as the work by Jokar et al. (2015), focuses on random-based, mean-based, and reverse-time-based reduction methods to differentiate between benign and attack data in binary classification models.
- **Need for Realism:** There is a gap in simulating authentic electricity theft patterns that closely mimic real-world complexities.
- **Validation with Real Attacks:** Existing studies lack rigorous validation of synthetic binary classification models against actual attack data to identify anomalies or outliers.
- **Complexity in Simulation:** The current literature does not fully address the challenge of creating sophisticated simulations that accurately represent the multifaceted nature of electricity theft.
- **Enhanced Anomaly Prediction:** A research opportunity exists to improve anomaly detection methods by validating models with genuine attack instances.

To address these gaps, the proposed research:

- **Incorporates Complex Theft Patterns:** Aims to integrate more intricate and realistic patterns of electricity theft into the training process.
- **Validates with Real Data:** Seeks to validate the effectiveness of trained models in predicting anomalies by testing them against verified real attack datasets.
- **Advances Methodologies:** Introduces more advanced methodologies that go beyond simple reduction techniques to provide a nuanced understanding of electricity theft.
- **Contributes to Anomaly Detection:** Enhances the body of knowledge in anomaly detection with the potential application in smart grid cybersecurity.

(d) Our Research Contributions are:

- **Cost-Efficiency:** Training ETD models with synthetic attack data bypasses the need for gathering real-life attack patterns, saving time and resources.
- **Real-World Evaluation:** Enables testing of ETD models against synthetic attacks and real data, ensuring cost-effective and practical assessment.
- **Simplified Calculation:** Streamlines the process of incorporating attack scenarios into meter readings by adding synthetic data to legitimate usage.
- **Previously Unattainable:** Offers a solution to the challenge in computer network communications where node behavior analysis depended on actual attack occurrences.
- **Enhanced Flexibility:** Synthetic data's adaptability, informed by a wide knowledge base, can significantly improve ETD model training.
- **Improved Training and Precision:** Leverages the diversity of synthetic attacks to train more robust ETD models, leading to accurate attack detection and classification in smart homes

Author action: We updated the manuscript by creating Motivation, Literature review, study gaps and contribution subsections under Introduction.

Reviewer#3, Concern # 3 (please list here): 3. The structure of the paper as a whole is inconsistent with the overall review:

Author response: We have restructured the paper as according to the suggestion above for consistency

Author action: We updated the manuscript by re-arranging each section in our manuscript for consistency.

Reviewer#4, Concern # 1 (please list here):

1) What is the rationale behind classifying the attacks to 5 categories shown in Fig. 1. In other words, given that the proposed framework in Fig. 3 is able to recognize the attack and also the class of the attack, what the importance of such a detection/classification will be for the operator?

Author response: Thank you for taking your precious time to read and review our manuscript. We appreciate your valuable suggestions and have revised our paper accordingly. Below are the responses to your questions and concerns:

Comparing the detailed of multiclass classification in Figure 1 with the preprocessing for binary classification in Figure 2, we discussed several points regarding their implications within the context of an anomaly detection framework like the one shown in Figure 3:

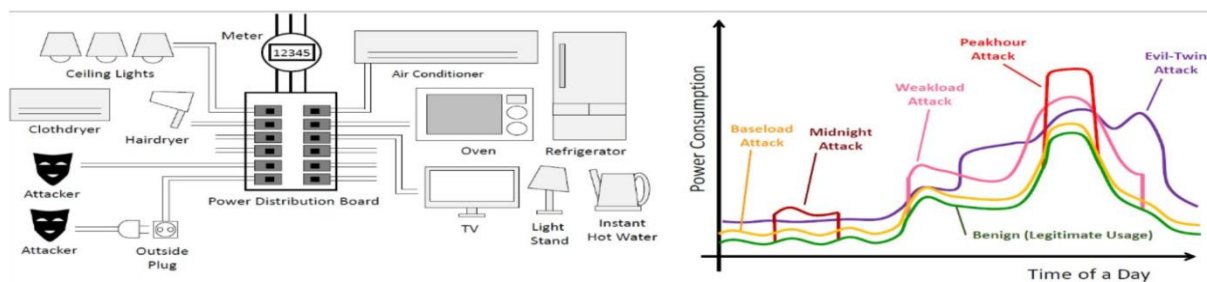


Fig. 1: Scenarios of electricity theft attacks in this study. Power distribution boards and cables are deployed behind walls and are usually not visible. An attacker uses the pre-deployed cable in a complex building or the outside plug to steal electricity. Depending on the theft pattern, we identify five classes of attacks.

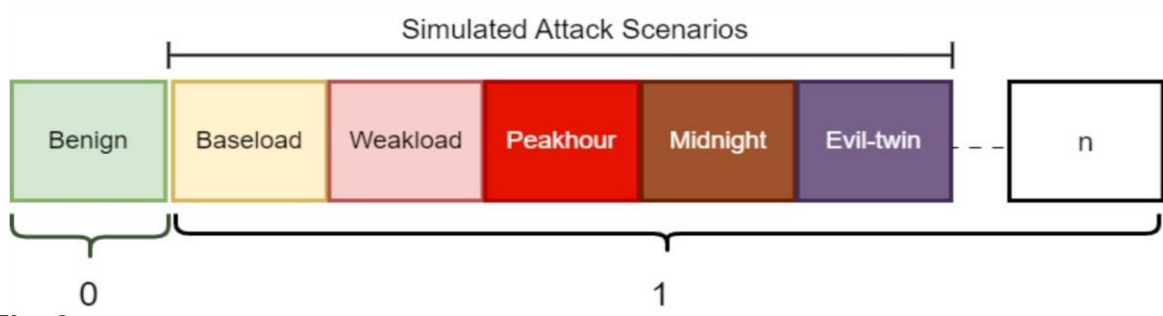


Fig. 2: Normalized range of data or a binary classification where '0' represents benign and '1' represents malicious or attack activities.

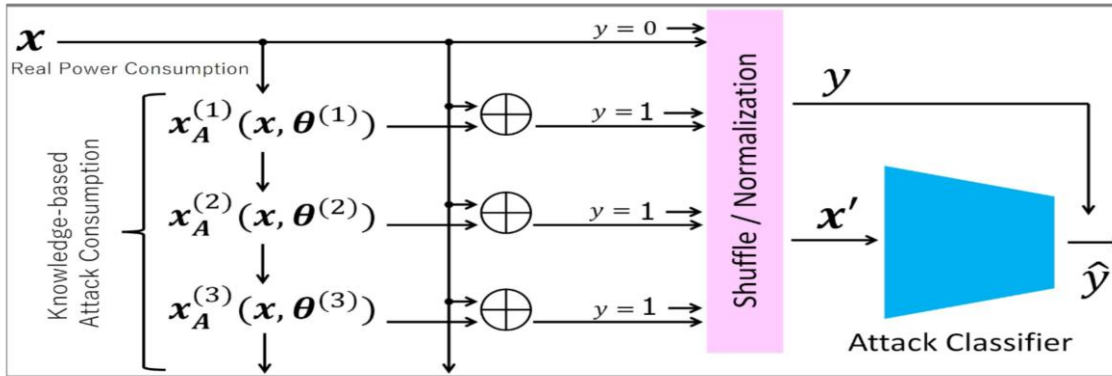


Fig. 3: The framework for electricity theft detection (ETD) processes real power consumption data to detect anomalies that indicate cyberattacks. It does so by transforming the data, labeling it, preparing it for analysis, and finally classifying it using a trained attack classifier

Explanation: The caption in Fig 3 was wrong and it has been corrected. Unlike in our previous paper [17] where we categorized smart homes attacks into 5 categories, this paper expands the limitations of the models to detect attacks that are not classified hence we preprocessed our dataset to accommodate the 5 attacks scenarios + unknown attacks, n , as '1' and the benign or non-attack data as '0' for our proposed ETD framework in Fig. 3 to detect known attacks and classify unknown as anomalies. The details of comparison between Multiclass classification (Fig. 1) and Binary Classification (Fig. 2) using different aspects such as Granularity, Complexity, Response Strategy, etc., are shown in the table below:

Aspect	Multiclass Classification (Fig. 1)	Binary Classification (Fig. 2)
Granularity	Identifies specific types of attacks.	Distinguishes between normal and abnormal behavior.
Complexity	Requires complex models and more data.	Simpler models and potentially less data needed.
Response Strategy	Allows for tailored responses to specific attacks.	General response to any detected anomaly.
Adaptability to Unknown Attacks	May not classify new attack types correctly.	Better at detecting new or unknown types of attacks.
System Overhead and False Positives	Higher system overhead and potential for more false positives.	Lower overhead and potentially fewer false positives.

Model Training and Evolution	Needs continuous updates for new attack types.	Easier to maintain and update.
Forensics and Reporting	Provides detailed information for in-depth analysis.	Limited information for detailed forensics.

The proposed framework in Figure 3 benefits from recognizing both the specific attacks and unclassified anomalies, integrating the strengths of both approaches for a robust security posture.

The detection and classification of cyber threats for operators are vital for several reasons:

- **Rapid Response:** Binary classification facilitates immediate threat recognition, enabling swift action to mitigate potential cyber threats.
- **In-Depth Analysis & Prevention:** Multiclass classification provides detailed insights into attack types, guiding resource allocation and the development of prevention strategies.
- **Strategic Planning & Resilience:** Specific attack classifications aid in tailored incident response planning and enhance overall system resilience.
- **Compliance & Confidence:** Adherence to regulatory standards through accurate detection and classification bolsters customer confidence in security measures.
- **Operational Integrity:** Efficient detection and classification are vital for maintaining operational continuity and minimizing disruptions from cyber incidents.

In essence, precise and swift identification of various cyberattack types is critical for operators to safeguard the integrity, availability, and confidentiality of smart home systems.

Author action: We updated the manuscript by

Reviewer#4, Concern # 2 (please list here): 2) Please provide more insights on the model selection and feature selection for the proposed machine learning framework. I recommend including more information about maximum-likelihood parameter estimation while optimizing the objective function.

Author response: (a) Model Selection:

XGBoost (SYN-XGB) demonstrates superior performance with highest AUC (98.69%) and ACC (95.54%) among the models. It's ensemble learning with boosting makes it robust to anomalies and noise in the electricity consumption dataset. Regularization in the objective function helps prevent overfitting, contributing to better generalization.

The Maximum Likelihood Parameter Estimation (MLE) in the context of our proposed (XGBoost) model selection, involves finding the values of hyperparameters, e.g, learning rate (eta), max-depth or number of trees(n_estimators) that maximize the likelihood of observing the actual data. After we performed Grid Search with Cross-Validation, the objective function, which is a combination of a loss function and a regularization term (L1 and L2) with values 0.01 and 1 respectively was minimized during training, selection was made based on the hyperparameter tuning of each model after the cross validation.

XGBoost employs a gradient boosting framework, and during each iteration, it fits a weak learner to the negative gradient of the loss function.

Regularization terms (L1 and L2 regularization) are included to prevent overfitting.

Table 7 has been updated with MLE from grid search cross validation in optimizing the objective function in determining our model selection.

About the features of Appliance consumption Dataset

Feature Selection for the proposed ML framework:

With reference to section V of our previous paper [17], we have used AMPds2 (Almanac of Minutely Power Dataset version 2) [20] as a benchmark for the two-year power consumption of homes. AMPds2 consists of minute-level measured powers at the outputs of a power distribution board. The names of the appliances monitoring points are shown in Table I.

The main features for our proposed machine learning framework is the aggregated power consumption data (**power_base**). Each entry in **power_base[day][minute]** represents the aggregated power consumption for a specific category at a given day and minute. The data is structured in a way that each day has 1440 minutes, and for each minute, there is aggregated power consumption information for different categories.

Features (Independent Variables):

1. Aggregated Power Consumption Data (**power_base**):

- **Structure:** The **power_base** dictionary contains aggregated electricity consumption data for different days and minutes.
- **Representation:** Each entry in **power_base[day][minute]** represents the aggregated power consumption for a specific category at a given day and minute.

Proposed XGB Machine Learning Framework:

Input Features (X):

- **Shape:** The input feature matrix (**X**) would have dimensions (730, 1440 * num_categories), where 730 is the number of days and num_categories (e.g., len(HPE)).
- **Flattening:** Flatten the nested structure of **power_base** to create a feature matrix where each row represents a day, and each column represents the aggregated power consumption for a specific category at a given minute.

Output Labels (y):

Labels: The output labels (**y**) correspond to specific conditions related to power consumption for the categories in **target_names**.

Feature Importance:

- XGBoost provides a feature importance score based on the contribution of each feature to the model's predictions.
- Feature importance can be analyzed to understand which categories and times of day are most influential in predicting the conditions related to power consumption.

Prediction:

- After training, our proposed model (XGBoost) is used to make predictions on new data.

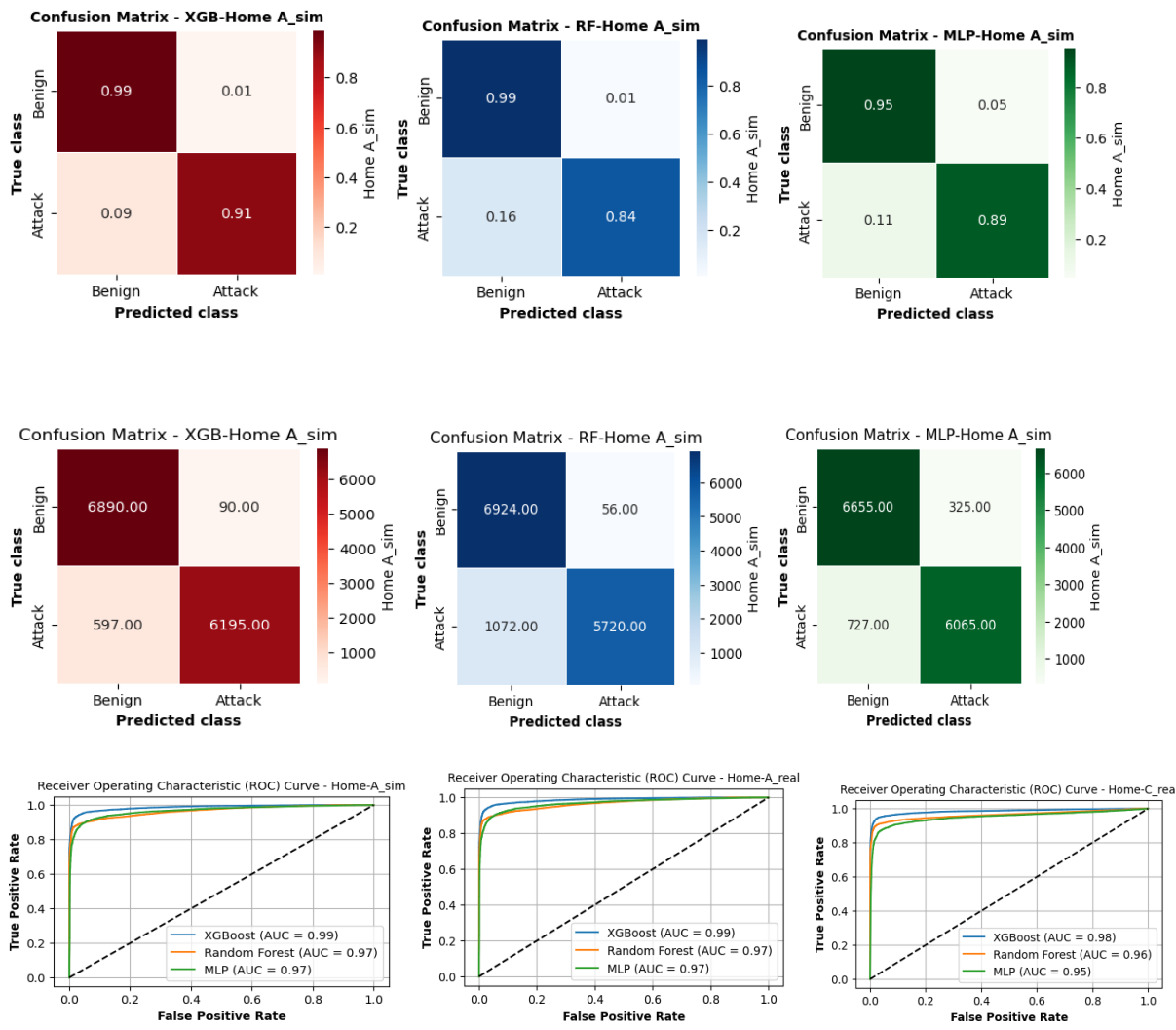
The aggregated power consumption data (**power_base**) serves as the primary feature for the proposed (XGBoost) machine learning framework. The model is trained to predict specific conditions related to power

consumption for the target categories (**target_names**). The maximum-likelihood parameter estimation involves optimizing an objective function that combines a loss function and regularization terms. The hyperparameters are tuned to optimize model performance, and feature importance analysis provides insights into the influential factors in the predictions.

Author action: We updated the manuscript by elaborating more about model and feature selection of our proposed ML framework with relevant references.

Reviewer#4, Concern # 3 (please list here): 3) Please fill the gap between confusion matrices and receiver operating characteristic curves. It is not clear how the curves represent a general view on the content of the confusion matrices. Please kindly explain.

Author response: We randomly selected 3 ROC curves and confusion matrices to explain our models from 18 confusion matrices for all our SYN models – XGB, RF and MLP. Although the MLP in Figure 12 was not properly represented. We corrected this error in our updated manuscript as shown in the figures below. Also, the confusion matrices in our manuscript were normalized, the un-normalized version are shared in the explanation below. We selected Home A_sim ROC curve and Confusion matrices for each of the model. The same principle applies to the other synthetic models.



The ROC curve provides a general view of the model's performance across all thresholds, giving us a sense of its discrimination ability. In contrast, a confusion matrix provides detailed information about the performance of our model at a specific threshold level.

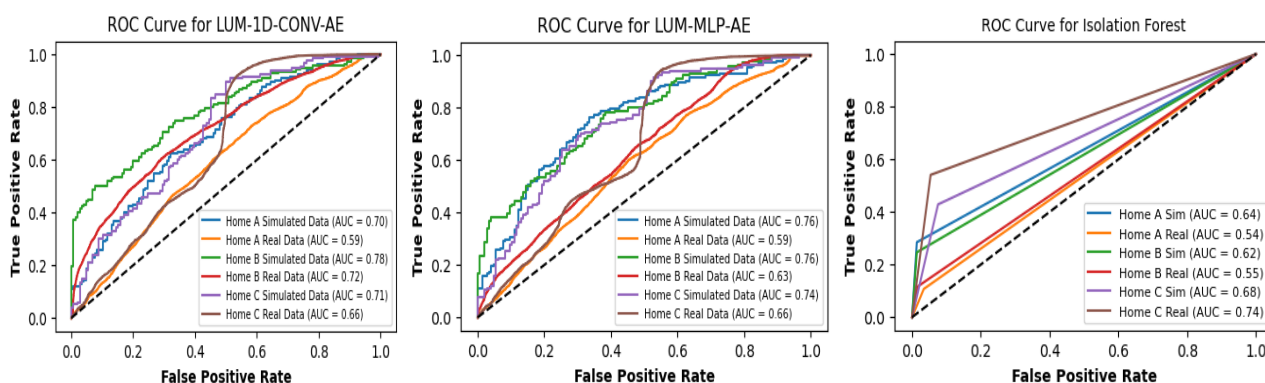
In the ROC curve, the AUC for each model (XGBoost, Random Forest, MLP) gives a single measure of performance across all possible classification thresholds, summarizing the trade-off between the TPR and FPR.

The confusion matrices give a more granular view. For instance:

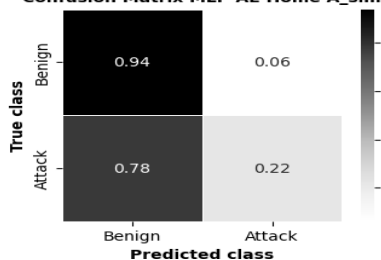
- The MLP confusion matrix indicates that the model correctly identifies 95% of benign cases (TN) and 89% of attack cases (TP) at a specific threshold.
- The Random Forest confusion matrix shows a high TN rate of 99% but a lower TP rate of 84%.
- The XGBoost confusion matrix shows a similar high TN rate of 99% and a better TP rate of 91%.

The ROC curve doesn't show the actual values of TP, FP, TN, and FN; it rather shows the rate at which these values change with different thresholds. A high AUC reflects a model that has a high TPR and a low FPR across different thresholds, which generally corresponds to high values of TP and TN and low values of FP and FN in the confusion matrices at a particular operating threshold.

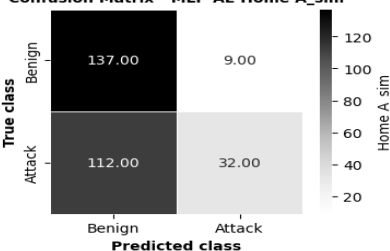
LEGACY UNSUPERVISED MODELS (LUM):



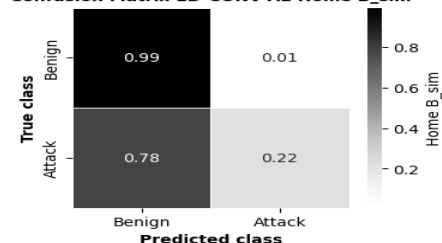
Confusion Matrix MLP-AE Home A_sim



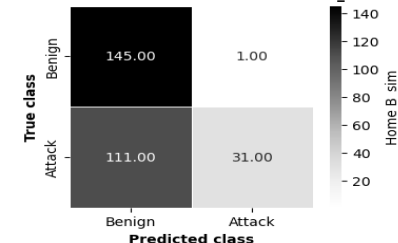
Confusion Matrix - MLP-AE Home A_sim

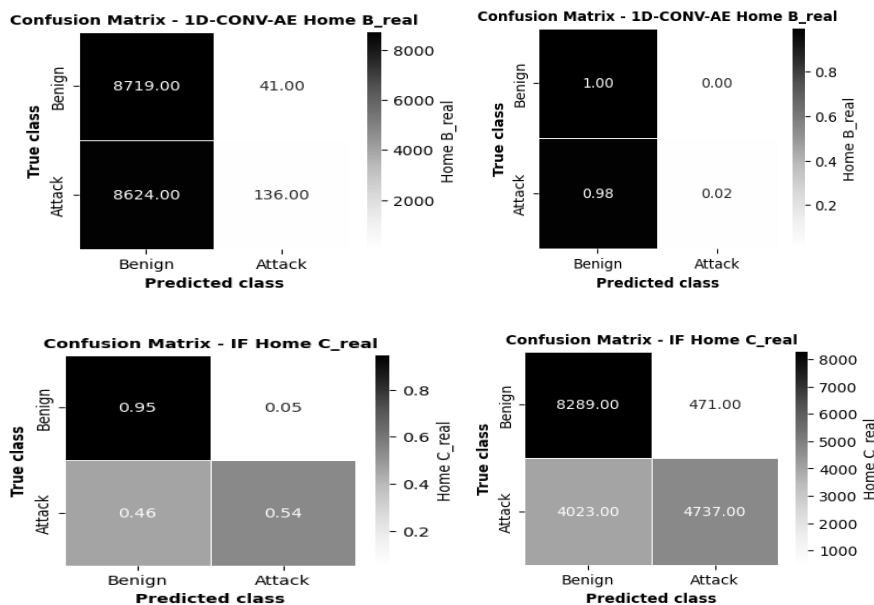


Confusion Matrix 1D-CONV-AE Home B_sim



Confusion Matrix - 1D-CONV-AE Home B_sim





From the confusion matrix, you can calculate various performance metrics such as precision, recall, specificity, and F1 score at a specific operating point (threshold).

When you look at the ROC curve, each point on the ROC curve represents a different threshold applied to the classifier to decide between the positive and negative class. Each of these points corresponds to a different confusion matrix with its own set of TP, TN, FP, and FN.

For instance, if we consider the confusion matrices for the simulated and real data from Home B with the 1D-CONV-AE model:

- The AUC of 0.78 and 0.72 for simulated and real data respectively on the ROC curve suggests that the model's ability to distinguish between the classes is reasonably good for simulated data and less so for real data.
- Looking at the corresponding confusion matrices, we see high TN rates (0.99 for simulated, 1.00 for real) but varying TP rates (0.78 for simulated, 0.98 for real). This suggests that while the model is quite good at identifying negative cases (benign), its performance on positive cases (attacks) is inconsistent between the simulated and real data.
- In the confusion matrices, we see the specific number of instances that are correctly and incorrectly classified, which is reflected in the ROC curve by how close the curve follows the top left corner.

From the ROC curve for the Isolation Forest model:

- **Home A Simulated** has an AUC of 0.64, meaning the model has a 64% chance of correctly distinguishing between a benign and an attack instance for the simulated environment of Home A.
- **Home A Real** has a lower AUC of 0.54, suggesting the model is less effective at distinguishing between benign and attack instances in the real-world data of Home A.

Similarly, **Home B Simulated** and **Home B Real** have AUCs of 0.62 and 0.55, respectively, and **Home C Simulated** and **Home C Real** have AUCs of 0.68 and 0.74.

The higher the AUC, the better the model is at distinguishing between the positive class (attacks) and the negative class (benign). For example, the model performs best on the real data for Home C, with an AUC of 0.74.

Confusion Matrix Details

From the confusion matrix for Home C real data:

- **True Positives (TP):** 4737 - The model correctly identified 4737 instances of attacks.
- **True Negatives (TN):** 8289 - The model correctly identified 8289 instances as benign.
- **False Positives (FP):** 471 - The model incorrectly identified 471 benign instances as attacks.
- **False Negatives (FN):** 4023 - The model failed to identify 4023 attacks, mistakenly

Relating the Confusion Matrix to the ROC Curve

- The specific values in the confusion matrix correspond to a single point on the ROC curve for Home C real data. The point is determined by the TPR (sensitivity) and FPR.

These values would find a corresponding point on the ROC curve, but that exact point isn't marked in the ROC curve. However, we know that point exists, and if the threshold were adjusted, this point would move along the curve, which would also result in different values in the confusion matrix.

In summary, the ROC curve tells us how well the model can separate the two classes and gives a holistic view of the model's performance across all thresholds. In contrast, the confusion matrix tells us where exactly the model is making mistakes at a specific threshold.

Author action: We updated the manuscript by adding more figures and table to validate reviewers' comments

Reviewer#4, Concern # 4 (please list here): 4) Please report the training error and test error of the proposed framework. How to guarantee a trade-off between these two errors?

Author response: (a) ML Supervised Learning

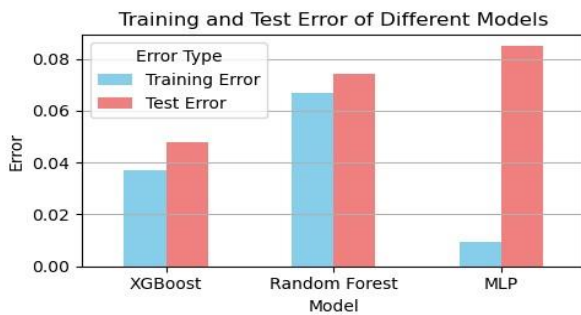
For classification tasks, the errors are often reported as:

- **Training Error:** 1-accuracy on the training set
- **Test Error:** 1-accuracy on the test set

The accuracy on each set is calculated as the number of correct predictions divided by the total number of predictions made.

This table presents the training and test errors for the XGBoost, Random Forest, and MLP models, respectively.

Metric	XGB (Proposed)	RF	MLP	SVM	LR
ACC	0.9554	0.9275	0.9331	0.9035	0.8148
Recall	0.9213	0.8581	0.9472	0.8432	0.7311
Precision	0.9882	0.9916	0.9399	0.9471	0.8565
AUC	0.9869	0.9676	0.9632	0.943	0.8748



To guarantee a trade-off between training and test errors, we use GridSearchCV for hyperparameter tuning and include cross-validation(cv5) for all the model, regularization parameters L1(reg_alpha) and L2(reg_lambda) for XGBoost. For Random Forest, the parameters **min_samples_split** and **min_samples_leaf** act as a form of regularization by preventing the tree from growing too deep or complex. For MLP, **alpha** is used as an L2 penalty term and the **max_iter** parameter is set, which combined with the default **early_stopping=True** setting.

The table presents the test AUC for the XGBoost, Random Forest, and MLP models, respectively. These results were obtained using hyperparameter tuning with GridSearchCV, cross-validation, and regularization as specified for each model.

Model	Test AUC
XGBoost	0.984347
Random Forest	0.966294
MLP	0.954237

(b) Unsupervised Learning Proposed Framework

Training Error

The training error in an autoencoder is often measured as the reconstruction error, which is the difference between the input data and the reconstructed data output by the autoencoder. In the code, the training error is calculated as the mean squared error across all features for each data point:

```
train_predictions = autoencoder.predict(X_train_scaled)
```

```
reconstruction_error = np.mean(np.square(X_train_scaled - train_predictions), axis=1)
```

Test Error

The test error is calculated similarly for the test dataset, which the model has not seen during training:

```
test_predictions = autoencoder.predict(X_test_scaled)
```

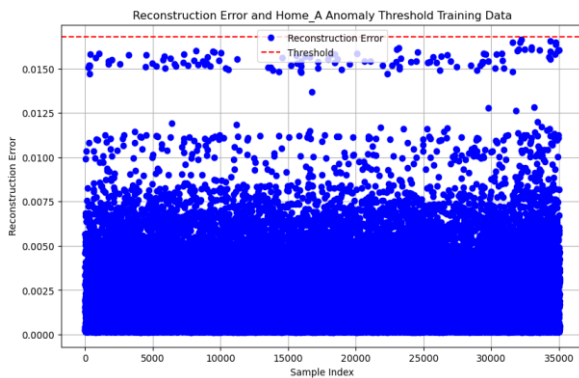
```
test_reconstruction_error = np.mean(np.square(X_test_scaled - test_predictions), axis=1)
```

Reporting Training and Test Errors

We typically calculate and display the average or median reconstruction error for the training and test datasets. The threshold for anomaly detection is then applied to classify whether a test sample is normal or an anomaly, which is used to calculate various performance metrics.

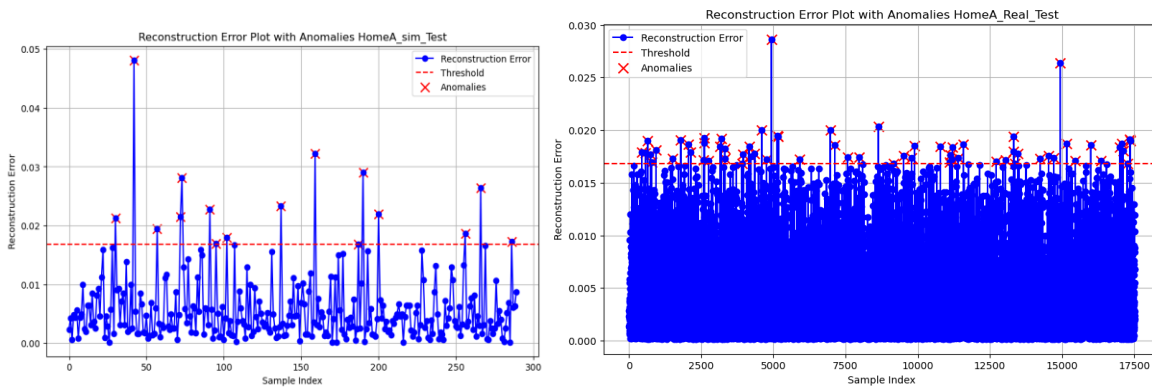
Visualizing Errors and Performance

The figures below show the plots for visualization of the reconstruction errors for anomaly threshold determination. The reconstruction error plot shows the threshold line, above which data points are



considered anomalies. The ROC curve provides insight into the trade-off between true positive rate and false positive rate at various threshold settings, with the AUC score summarizing the model's overall capability in anomaly detection.

Fig. 1: Determining the Threshold of our Autoencoder using the training Data.



Trade-off Between Training and Test Errors

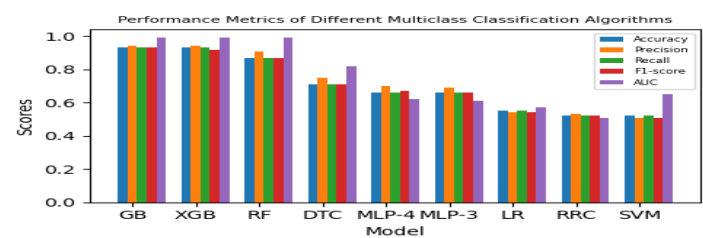
In machine learning, particularly in unsupervised anomaly detection with autoencoders, we aim to minimize the reconstruction error on both the training and test datasets. However, too low a training error (overfitting) may result in poor generalization to the test data. The following methods help us achieved a good trade-off:

1. **Early Stopping:** The code uses an **EarlyStopping** callback to stop training once the validation loss stops improving, which helps prevent overfitting.
2. **Validation Split:** By setting aside a portion of the training data for validation (as in **validation_split=0.1**), the model's generalization capabilities are monitored during training.
3. **Regularization:** Although not explicitly used in the code, techniques such as L1 or L2 regularization can be applied to the layers of the autoencoder to penalize overly complex models.
4. **Hyperparameter Tuning:** Adjusting the size of the encoding (32,64,) and hidden layers (16, 32,) and the training hyperparameters (such as learning rate) help us find a balance between underfitting and overfitting.
5. **Threshold Setting:** The threshold for classifying an anomaly is set based on the training data reconstruction error, with some multiplier of the median absolute deviation. Adjusting this threshold helps us balance sensitivity and specificity.

Author action: We updated the manuscript by adding more figures and references.

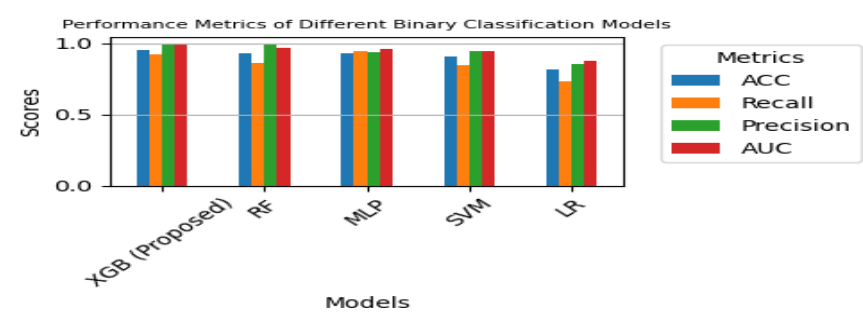
Reviewer#4, Concern # 5 (please list here): 5) The performance of the proposed framework needs to be compared with other machine learning-based techniques (e., SVM) available in the existing literature. Please also report the limitations of the proposed framework.

Author response: **Author response:** In our previous paper [17] we evaluated our models with 9 different classifiers including SVM as shown in the below:



However, ensemble methods like GB and XGB demonstrate the best overall performance for the multi-class ETD task, with high scores in all evaluated metrics. The precision-focused RF algorithm and the more nuanced MLP-4 also show promising results. In contrast, traditional algorithms like SVM, LR, and RRC appear less effective for this particular application.

We further carried out experiment with SVM and LR binary classifiers, our proposed XGB framework demonstrates the most balanced performance and is likely the most suitable model for ETD among the evaluated options as visualized below:



Our the proposed XGB model has the highest scores in ACC and AUC as shown in table below, indicating its superior overall performance among the evaluated models. The RF model shows remarkable Precision, while the MLP has the highest Recall, which could make it the most sensitive model to detecting the positive class. The SVM and LR models trail behind the others across most metrics, with LR having the lowest scores in all categories as shown in the table below:

Metric	XGB (Proposed)	RF	MLP	SVM	LR
ACC	0.9554	0.9275	0.9331	0.9035	0.8148
Recall	0.9213	0.8581	0.9472	0.8432	0.7311
Precision	0.9882	0.9916	0.9399	0.9471	0.8565
AUC	0.9869	0.9676	0.9632	0.943	0.8748

Limitations of our proposed framework

1. Reliance on Synthetic Data: The model primarily uses synthetic attack data, which may not fully capture the complexity and variability of real-world scenarios. There's an ongoing debate about the extent to which this synthetic data can accurately represent real-world conditions.

2. **Limited Real-World Attack Data:** The model has been tested with real-world attack data, but the amount of this data is limited. Expanding the dataset to include a broader and more diverse range of real attack scenarios is necessary to enhance the model's applicability and reliability in real-world contexts.
3. **Generalizability Concerns:** The model's effectiveness has been evaluated in specific home environments (Home A, B, and C). However, smart homes vary widely in their configurations, raising questions about the model's generalizability to different types of smart homes.
4. **Data Privacy Challenges:** With increasing concerns about data privacy, it's essential to ensure that data collection methods, especially those involving real-world attack data, adhere to privacy standards. Solutions like federated learning or secure multiparty computation could help in detecting electricity theft in smart homes without compromising customer privacy.

Author action: We updated the manuscript by removing our model limitations from Discussion and Future work section and create a subsection for our model limitations.

Reviewer#4, Concern # 6 (please list here): 6) There are several typos throughout the draft (e.g., Table ??, etc.). Please do a second, more focused, proofreading before final decision can be made.

Author response: Thank you for your comments, we reviewed and revised our paper/manuscript thoroughly. We proofread and correct the typos throughout the manuscript.

Author action: We updated the manuscript by checking with PaperPal Preflight to check for grammar issues.

Reviewer#4, Concern # 7 (please list here): 7) Is the subject matter presented in a comprehensive manner? No. It is not.

Author response: We have updated all the comments from the reviewers and represented our manuscript according IEEE Access Journal guidelines.

Author action: We updated the manuscript by reviewing the reviewers' comments and IEEE Access Journal guidelines.

Note: *References suggested by reviewers should only be added if it is relevant to the article and makes it more complete. Excessive cases of recommending non-relevant articles should be reported to ieeeaccess@ieee.org*